

- 4 (a) e.g. storing energy
separating charge
blocking d.c.
producing electrical oscillations
tuning circuits
smoothing
preventing sparks
timing circuits
(any two sensible suggestions, 1 each, max 2) B2 [2]
- (b) (i) $-Q$ (induced) on opposite plate of C_1 B1
by charge conservation, charges are $-Q, +Q, -Q, +Q, -Q$ B1 [2]
- (ii) total p.d. $V = V_1 + V_2 + V_3$ B1
 $Q/C = Q/C_1 + Q/C_2 + Q/C_3$ B1
 $1/C = 1/C_1 + 1/C_2 + 1/C_3$ A0 [2]
- (c) (i) energy = $\frac{1}{2}CV^2$ or energy = $\frac{1}{2}QV$ and $C = Q/V$ C1
= $\frac{1}{2} \times 12 \times 10^{-6} \times 9.0^2$
= $4.9 \times 10^{-4} \text{ J}$ A1 [2]
- (ii) energy dissipated in (resistance of) wire/as a spark B1 [1]

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5 (a) supply connected correctly (to left & right)

P1